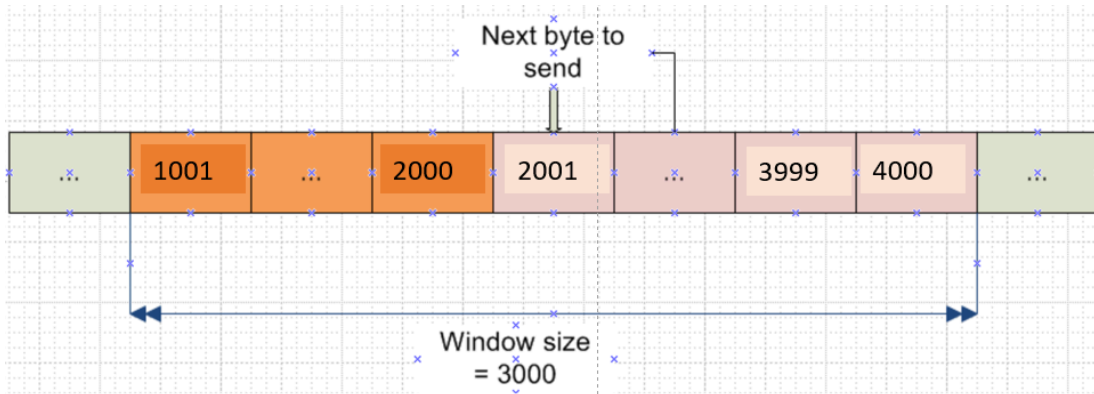
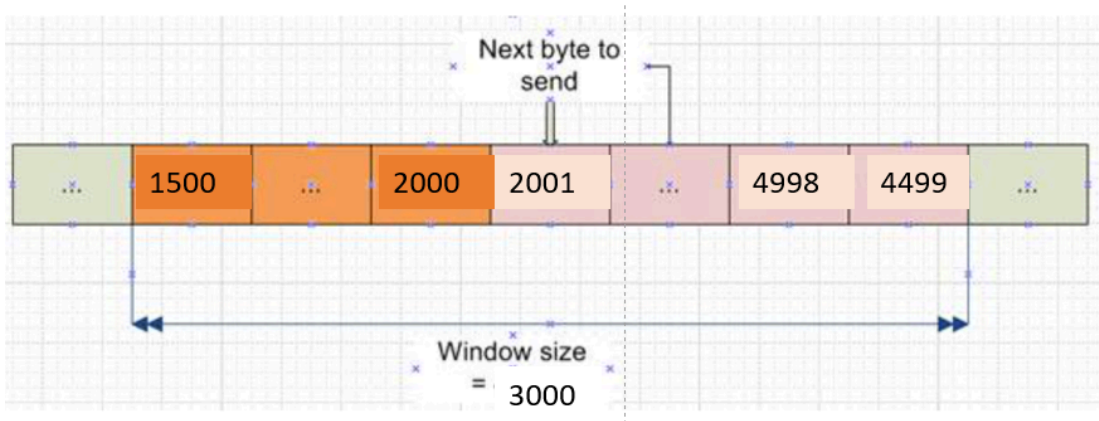


hw3 sol

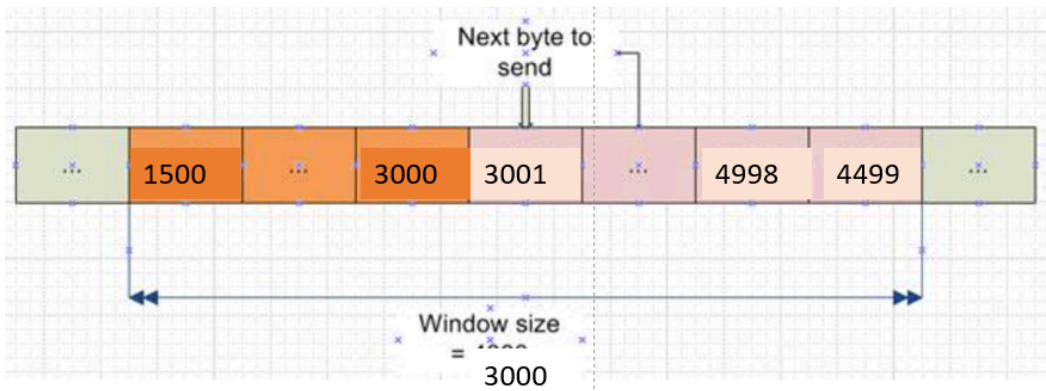
1. A window holds bytes 1001 to 4000. The next byte to be sent is 2001. Draw a figure to show the situation of the window after the following two events:



a. An ACK segment with the acknowledgment number 1500 and window size advertisement 3000 is received.



b. After a. A segment carrying 1,000 bytes is sent



c. After b, how many bytes can A send without waiting for new ACKs?

$$4499 - 3001 + 1 = 1499$$

2. Consider hosts A and B communicating over a TCP connection. Assume unrealistically that the initial sequence number of A and B is 3000 and 5000 respectively. Assume that all segments sent between A and B have 20-byte headers. Give the value of the sequence number field and acknowledgment number fields for each segment sent between A and B.

1) A sends B a segment with a 200 byte payload

Seq= 3001 Ack = 5001

2) B responds with a segment with a 100 byte payload.

Seq= 5001 Ack = 3201

3) After A receives the acknowledge segment from B, A responds with a segment with a 100-byte payload. The segment is lost

Seq= 3201 Ack = 5101

4) Right after 3), A response with another segment with a 250-byte payload.

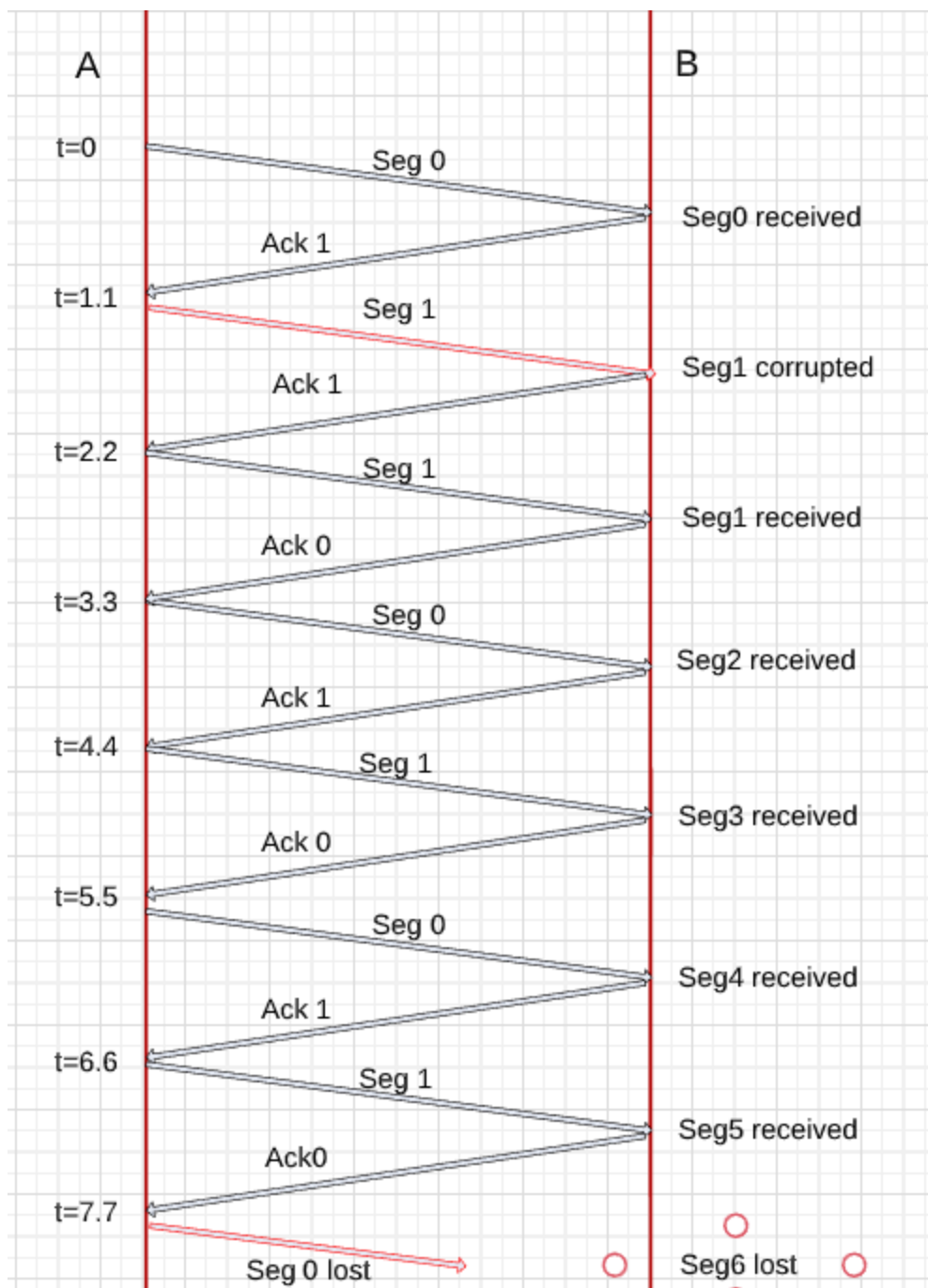
Seq= 3301 Ack = 5101

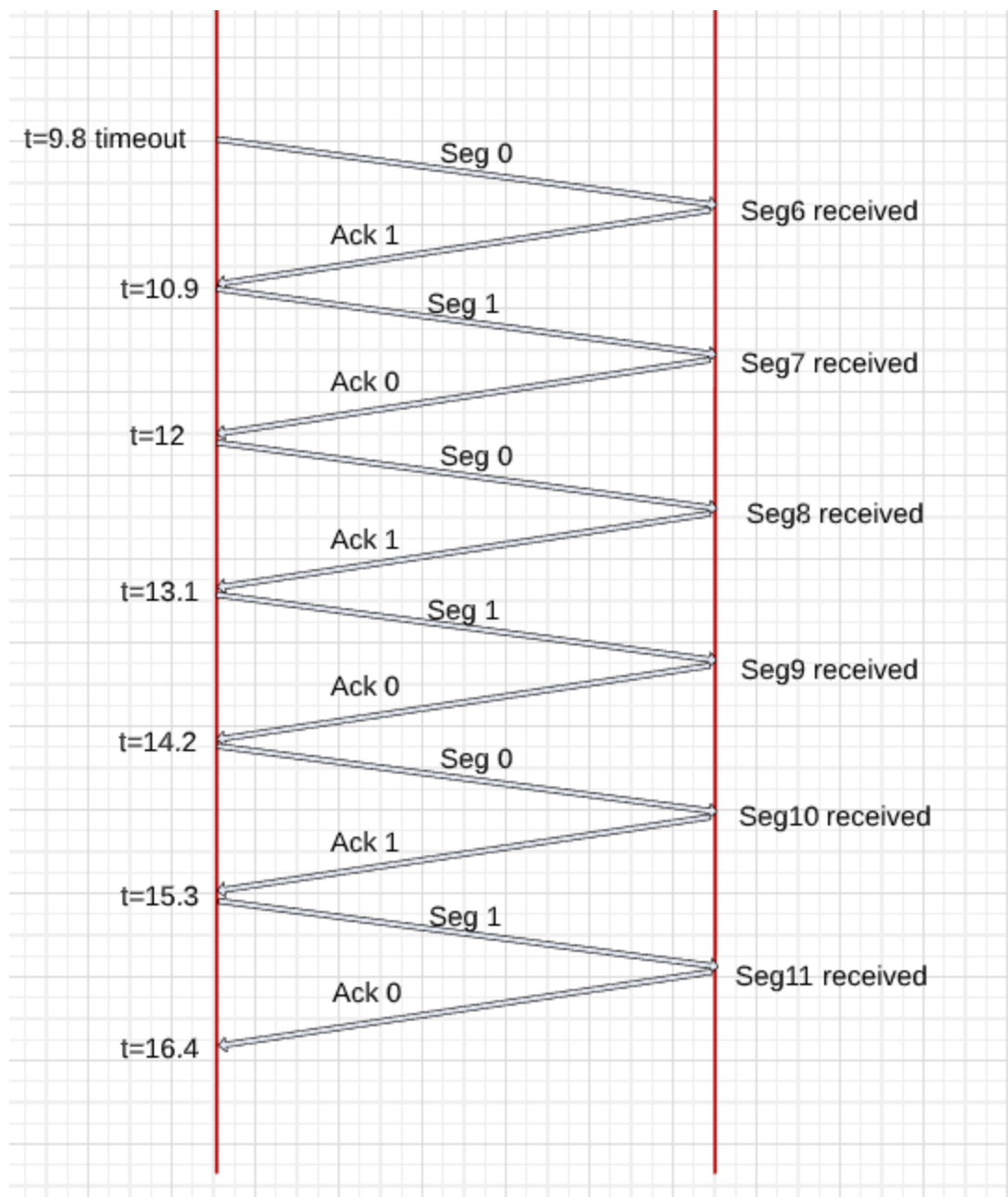
5) B responds to the segment sent in 4) with a 100-byte payload segment

Seq= 5101 Ack = 3201

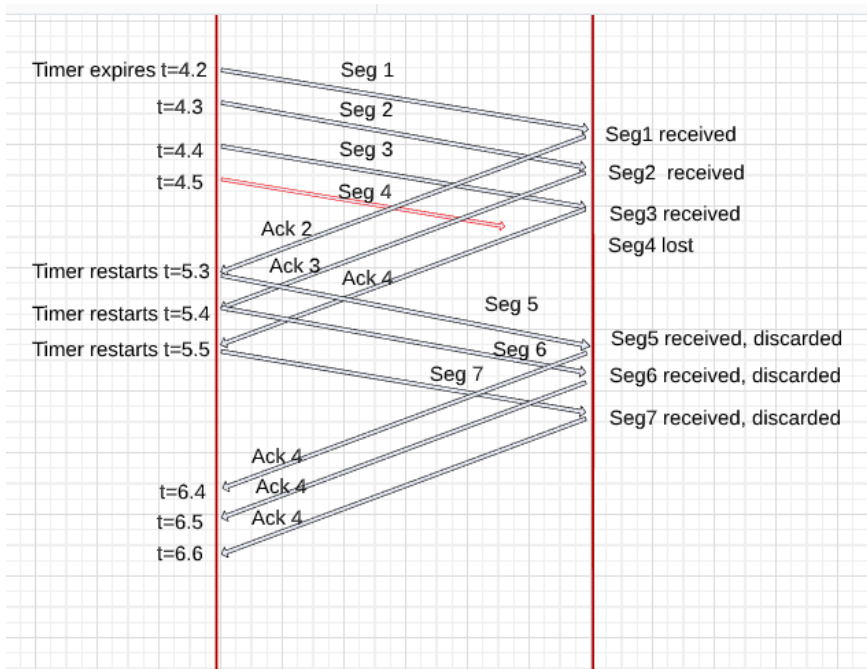
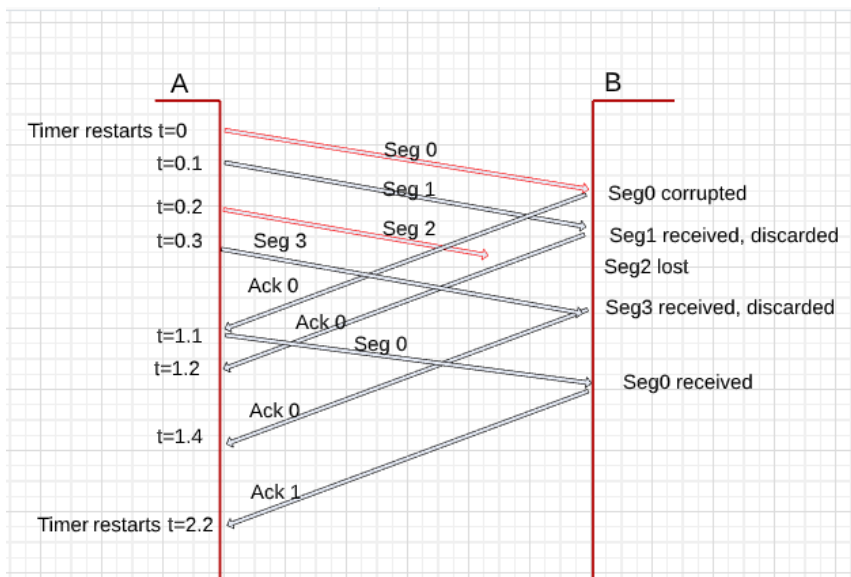
3. Consider a sender sends 12 segments to a receiver. The RTT between the sender and the receiver is 1 ms. The transmission time of each frame is 0.1 ms. The timeout value is 2 ms. In each of the following cases, how long does it take for the sender to transfer the 12 segments? (time between sending out the first segment and receiving the ack of the last segment). For each case, draw a diagram that shows the interaction between the sender and the receiver. You also need to provide the information about which segment is sent and the ACK number.

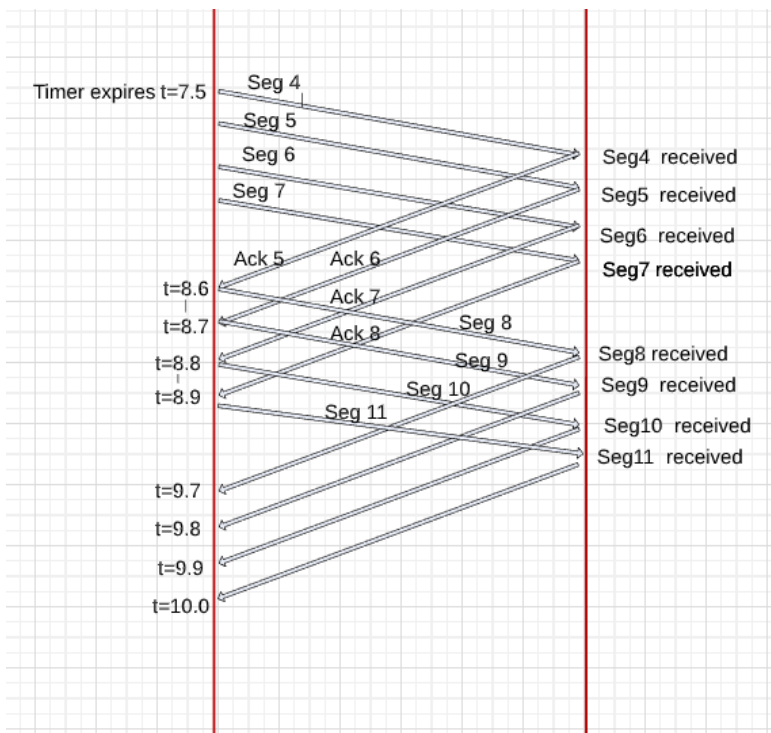
a. Stop-and-wait protocol V2 is used. The 2nd and the 6th segments are corrupted. The 6th segment is lost.





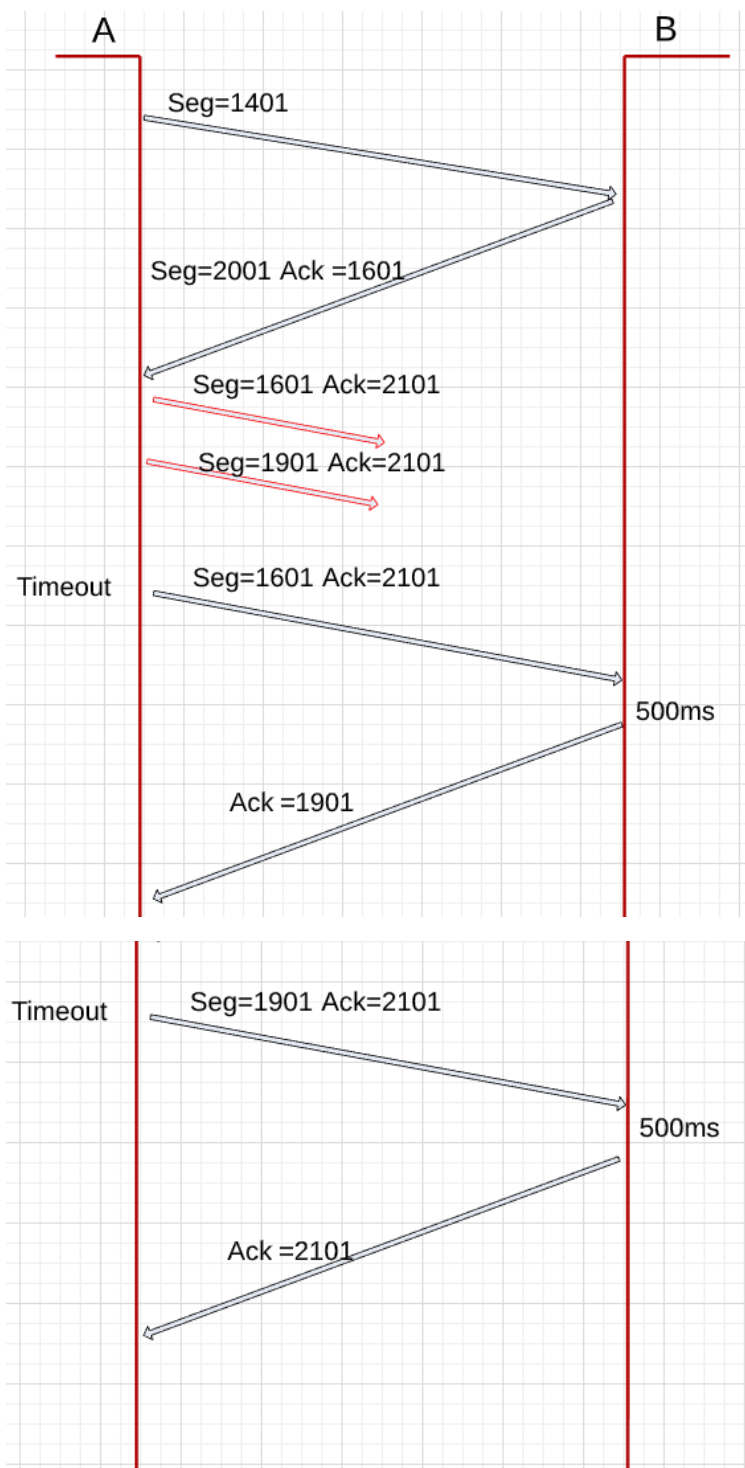
b. Go-Back-N protocol is used. The window size is 4. The 1st segment is corrupted. The 3rd and the 5th segments are lost during the first transmission





4. Show a timeline diagram for the following scenario. Specify the sequence number and ack number of each segment.

- A sends a segment carrying bytes 1401 to 1600, which arrives at B
- B sends a segment carrying bytes 2001 to 2100 and acknowledging the 1st segment from A, which arrives.
- A sends a segment carrying bytes 1601 to 1900 and acknowledging the segment arrived, but the segment is lost.
- A sends a segment carrying bytes 1901 to 2100, but the segment is lost
- Timeout occurs at A
- A resends a segment in response to time-out, this segment arrived.
- B sends an acknowledgment after the ACK-delaying timer expires.
- Another time-out occurs at A
- A resends a segment in response to time-out, this segment arrived.
- B sends an acknowledgment after the ACK-delaying timer expires.



5. A host sends 5 packets and receives 3 ACKs. The time is shown as hour:minute: second

Calculate the values of RTTM, RTTS, RTTD, and RTTO if the original RTO is 10 seconds. Did the send miss the retransmission of any segment? Show which segments should have been retransmitted. Rewrite the events including the retransmission time

a) Segment 1 was sent at 0:0:00

- b) Segment 2 was sent at 0:0:05
- c) ACK for segment 1 and 2 received at 0:0:07

- $RTT_M = 7$
- $RTT_S = 7$
- $RTT_D = 7/2 = 3.5$
- $RTO = 7 + 4 * 3.5 = 21$

- d) Segment 3 was sent at 0:0:20

Timer restarts $RTO = 21$

- e) Segment 4 was sent at 0:0:22
- f) ACK for segment 3 and 4 received at 0:0:35

- $RTT_M = 15$
- $RTT_S = 7 * 7/8 + 15 * 1/8 = 8$
- $RTT_D = 3.5 * 3/4 + 1/4 * (15 - 8) = 4.375$
- $RTO = 8 + 4 * 4.375 = 25.5 \approx 26$

- g) Segment 5 was sent at 0:0:37

- h) Timeout occurs at 0:0:63 and segment 5 should be retransmitted. $RTO = 2 * 26 = 52$

- i) ACK for segment 5 received at 0:0:75

Don't measure RTT for the retransmitted segment.